

PART I: NUMBER SYSTEMS

WHAT IS A NUMBER SYSTEM?

SAAD BAIG – DIGITAL LOGIC DESIGN



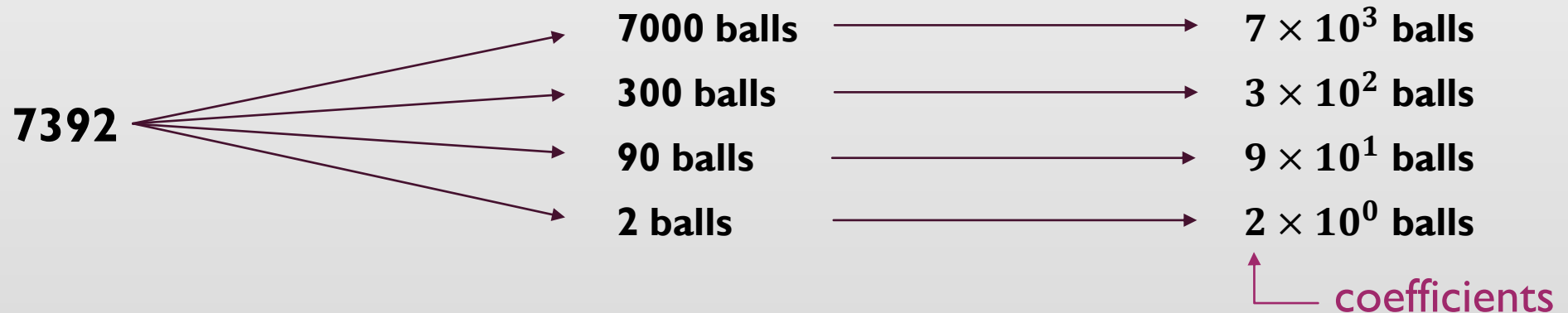
The animations and contents in these slides are originally adapted from the DLD course slides designed by Dr. Hasan Baig and Mr. Junaid Ahmed Memon. The original slides can be accessed from <https://www.hasanbaig.com/resources>

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LET'S START WITH THE DECIMAL SYSTEM



Visualize a ball pit with 7392 balls.





DECIMAL SYSTEM

- In decimal system, the convention is to write just the coefficients:

$$\dots a_3 a_2 a_1 a_0 . a_{-1} a_{-2} a_{-3} a_{-4} \dots$$

- Coefficients a_j are any 1 of the 10 digits (0 to 9)

- j defines the power of 10 by which the coefficients must be multiplied (10^j):

$$\dots 10^3 a_3 + 10^2 a_2 + 10 a_1 + a_0 + 10^{-1} a_{-1} + 10^{-2} a_{-2} + 10^{-3} a_{-3} \dots$$

- Since it uses 10 (0-9) digits and the coefficients are multiplied by a power of 10, decimal number system is called base 10, or radix 10.



NOW LET'S TALK ABOUT THE BINARY SYSTEM

- Since it's either a 0 or 1, there are only 2 possible values of coefficients a_j .
- Each coefficient is multiplied by a power of the radix (2^j). The results are added to obtain the decimal equivalent of the number.
- Example:

$$11010.11 \rightarrow 26.75$$

$$1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 + 1 \times 2^{-1} + 1 \times 2^{-2} = 26.75$$



NUMBER SYSTEMS

- In general, a number expressed in a base- r system has coefficients multiplied by powers of r :
$$a_n \cdot r^n + a_{n-1} \cdot r^{n-1} + \dots + a_2 \cdot r^2 + a_1 \cdot r^1 + a_0 + a_{-1} \cdot r^{-1} + a_{-2} \cdot r^{-2} + \dots + a_m \cdot r^m$$
- The coefficients a_j range in value from 0 to $r - 1$.
- Nomenclature: $(number)_{base}$

ACTIVITY

Create a number system from the last digit of your HU ID.



- (30 secs) Write down your HU ID, then write down the last digit of your HU ID that is not 0 or 2.
- (1 min) This number is your base or radix. Write down all coefficients that are associated with it.
- (1 min) Next write down the first 10 numbers of your number system.
- (2 min) Write down any 3 digit random number of your base using your coefficients and find out its value in decimal.
- (1 min) You will find that your number system exists out there! Do some research and find out its name.



NUMBER SYSTEMS EXAMPLE

- For a base-5 system (Quineral or Pental Number System):

- The coefficients for base 5 will range from 0 to 4.
- The first 10 numbers would be: 0, 1, 2, 3, 4, 10, 11, 12, 13, 14.

- **Example:**

$$(4021.2)_5 = 4 \times 5^3 + 0 \times 5^2 + 2 \times 5 + 1 + 2 \times 5^{-1} = (511.4)_{10}$$



OCTAL NUMBER SYSTEM

- Base 8 number system with coefficients from 0 to 7.
- The first 10 numbers would be 0, 1, 2, 3, 4, 5, 6, 7, 10, 11.
- Example:

$$(127.4)_8 = 1 \times 8^2 + 2 \times 8 + 7 + 4 \times 8^{-1} = (87.5)_{10}$$



HEXADECIMAL NUMBER SYSTEM

- Base 16 number system.
- The decimal number system only has 0 to 9 for representation. How would 16 coefficients be represented?
- Hexadecimal borrows alphabetical letters for coefficients.

Dec	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Hex	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F

- Example:

$$(B65F)_{16} = 11 \times 16^3 + 6 \times 16^2 + 5 \times 16 + 15 = (47687)_{10}$$



BINARY SYSTEM AND BITS

- The digits in a binary number are called **bits**.
- When a bit is equal to 0, it does not contribute to the sum during the conversion.
- Therefore, the conversion from binary to decimal can be obtained by adding only the numbers with powers of two corresponding to the bits that are equal to 1.
- **Example:**

$$(110101)_2 = 32 + 16 + 4 + 1 = (53)_{10}$$

- 8 bit equals one **byte**.



BINARY SYSTEM AND BITS

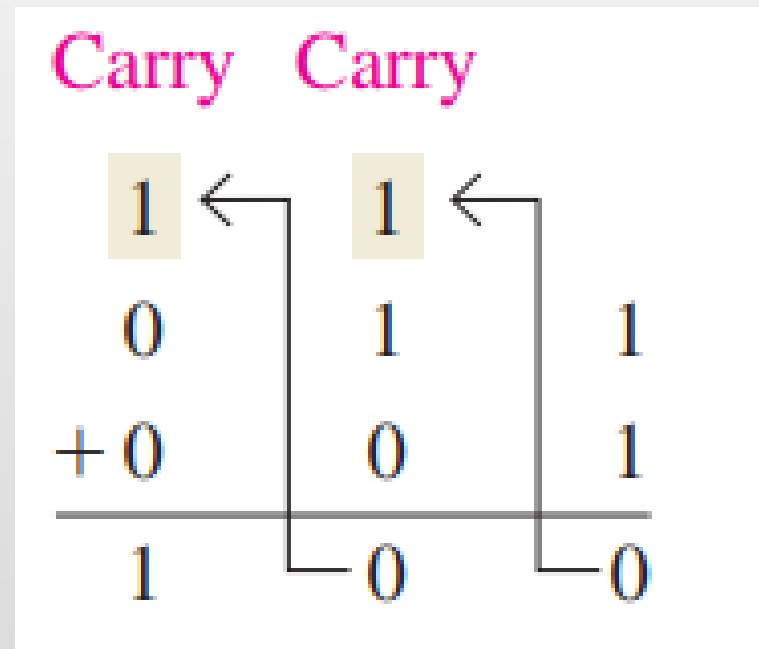
Powers of Two

n	2^n	n	2^n	n	2^n
0	1	8	256	16	65,536
1	2	9	512	17	131,072
2	4	10	1,024 (1K)	18	262,144
3	8	11	2,048	19	524,288
4	16	12	4,096 (4K)	20	1,048,576 (1M)
5	32	13	8,192	21	2,097,152
6	64	14	16,384	22	4,194,304
7	128	15	32,768	23	8,388,608

BIT ADDITION

- When a base other than base 10 is used, one must be careful to use only the r -allowable digits.

$0 + 0 = 0$	Sum of 0 with a carry of 0
$0 + 1 = 1$	Sum of 1 with a carry of 0
$1 + 0 = 1$	Sum of 1 with a carry of 0
$1 + 1 = 10$	Sum of 0 with a carry of 1



BIT SUBTRACTION

- When a base other than base 10 is used, one must be careful to use only the r -allowable digits.

$$0 - 0 = 0$$

$$1 - 1 = 0$$

$$1 - 0 = 1$$

$$10 - 1 = 1 \quad 0 - 1 \text{ with a borrow of } 1$$

Left column:

When a 1 is borrowed, a 0 is left, so $0 - 0 = 0$.

Middle column:

Borrow 1 from next column to the left, making a 10 in this column, then $10 - 1 = 1$.

Right column:

$1 - 1 = 0$

$$\begin{array}{r} 0101 \\ -011 \\ \hline 010 \end{array}$$

BIT MULTIPLICATION

- When a base other than base 10 is used, one must be careful to use only the r -allowable digits.

$$0 \times 0 = 0$$

$$0 \times 1 = 0$$

$$1 \times 0 = 0$$

$$1 \times 1 = 1$$

Partial products

$$\begin{array}{r} 111 \\ \times 101 \\ \hline 111 \\ 000 \\ +111 \\ \hline 100011 \end{array}$$



BIT DIVISION

- When a base other than base 10 is used, one must be careful to use only the r -allowable digits.
- Division in binary follows the same procedure as division in decimal

$$\begin{array}{r} 11 \\ 10 \overline{) 110} \\ \underline{10} \\ 10 \\ \underline{10} \\ 00 \end{array}$$



BIT OPERATIONS

augend:	101101	minuend:	101101	multiplicand:	1011
addend:	<u>+100111</u>	subtrahend:	<u>-100111</u>	multiplier:	<u>× 101</u>
sum:	1010100	difference:	000110		1011
				partial product:	0000
					1011
				product:	<u>110111</u>